order11 RESULTS: weight-1 completion of θ_E from the C_arc rays, and its obstruction
Prereg: PREREG.md, sha256 5d41a83a…d989, written before any numerics. Script hashes are in scripts/SHA256SUMS. Data: data/s.json. Log: logs/s.log.

Candidate
$\hat{F}_E = \theta_E + \tfrac{1}{2}(T_{c1} - T_{\gamma c1})$, where $T_{c1} = -\theta_{7/2}\cdot\Sigma_u \operatorname{sgn}(u)\cdot\operatorname{erfc}(\sqrt{(10\pi y)}|u|)\cdot e(-5u^2\tau/2 - 5b0\cdot u)$ is $\theta_{7/2}$ times the non-holomorphic Eichler integral of the $5n^2/2$ series.
$\xi_1\hat{F}_E = \tfrac{1}{2}\sqrt{10}\cdot y^{1/2}\cdot[\overline{\theta_{7/2}}\cdot\vartheta^{-}$ at (α,β) − the same at $(\gamma^{-1}\alpha, \gamma^{-1}\beta)]$.

Verdicts
Test	Result	Verdict
N1 parity: T is P-odd, θ_E is P-even (all 105 components)	9.9e−32 / 5.0e−32	pass
N2 ξ formula, by mp.diff	1.8e−31	pass (calculus check)
Control $\hat{G}_O$, $\chi = 1$ (dps 20)	3.6e−21	**pass**, so the test is valid
N3, $\chi = 1$, $\kappa = \tfrac{1}{2}$	0.68 (best-fit κ: 0.65)	**FAIL**
N3, $\chi = -1$, $\kappa = \tfrac{1}{2}$	0.69 (best-fit κ: 0.65)	**FAIL**
N3, $\chi = i$, $\kappa = \tfrac{1}{2}$	0.96 (best-fit κ: 0.92)	**FAIL**
N3, $\chi = -i$, $\kappa = \tfrac{1}{2}$	0.65 (best-fit κ: 0.53)	**FAIL**
θ_E alone, $\chi = \pm i$ (recheck of order8 M3 at dps 30)	0.92 / 0.53	**FAIL**

The best-fit κ is about 0.01–0.05, essentially 0: the correction cannot repair the law. $T_{c1} - T_{\gamma c1}$ alone also fails every χ (0.42–0.56).

Obstruction (Category A: Lemma O and its Corollary, PREREG §2)
- **Lemma O.** If a vector F of 105 functions, holomorphic or not, obeys $F(-1/\tau) = \chi(\tau/\sqrt{35})MF(\tau)$ for a scalar χ, then $F = -\chi^2 PF$. This follows from $M^2 = 35P$, which is exact.
- **Parity.** θ_E is P-even. Every Eichler correction built from weight-3/2 unary thetas paired with weight-½ thetas is P-odd.
- **No completion of this form exists** for any scalar χ and any κ:
  - $\chi = \pm 1$ forces the P-even part to vanish, which is impossible since $\theta_E \neq 0$.
  - $\chi = \pm i$ forces the P-odd correction to vanish, which reduces to θ_E (plus a holomorphic P-even term) obeying the $\chi = \pm i$ law. For θ_E alone this is falsified (Category B).
- The numbers above confirm this (Category B).
- Scope: this obstruction holds without holomorphy. The review's restriction concerns $|\tau|$-type laws, which are not of this form.

Full-series ξ-shadow
NOT located in any scalar-twisted fixed weight-1 M-law.
- The pieces already located (cone $\hat{H}$, odd pair $\hat{G}_O$) are P-odd, consistent with $\chi = 1$.
- The P-even sector (θ_E, and the even part of the full series) cannot carry a fixed weight-1 τ-law with $\chi = \pm 1$ at all.
- The even sector's natural law is the weight-0 / $|\tau|$-type law of $y^{1/2}\Phi_0$ (order7, order10).
- There remain three distinct unary-theta series ($5n^2/2$ shared). $\theta_E = \boxed{A}(\Phi)$ is an identification, not a bridge. $M^2 = 35P$ is exact (Category A).

Disclosures
- The S-law tests run at 3 interior τ in mpmath (dps 30; control at dps 20). These are fixed-matrix law tests, not float64 span tests, and they are secondary to Lemma O.
- Lattice boxes are truncated (tails below e^{-140}). θ_E for axis companions uses edge weight ½.
- The best-fit κ is descriptive only. Every verdict was recorded by the preregistered rule; no post-hoc rerun was done.
- The $\chi = \pm i$ route for $\theta_E + h$ with h holomorphic and P-even is not excluded in general; only $h = 0$ was tested.

Open next step
Two options:
(a) Test the $\chi = \pm i$ law for $\theta_E + h$, with h in the span of P-even holomorphic weight-1 theta series of the lattice (finite and exact: Hecke-type for the definite forms of discriminant $-35\cdot k$). Use exact q-expansion coefficients and solve for h by linear algebra.
(b) Formulate the P-even sector's law as a $|\tau|$-type weight-$(\tfrac{1}{2},\tfrac{1}{2})$ law and test $y^{1/2}\cdot$(even part) exactly. This is the review finding-1 type.